

# AOE 3044: Boundary Layer Theory and Measurement

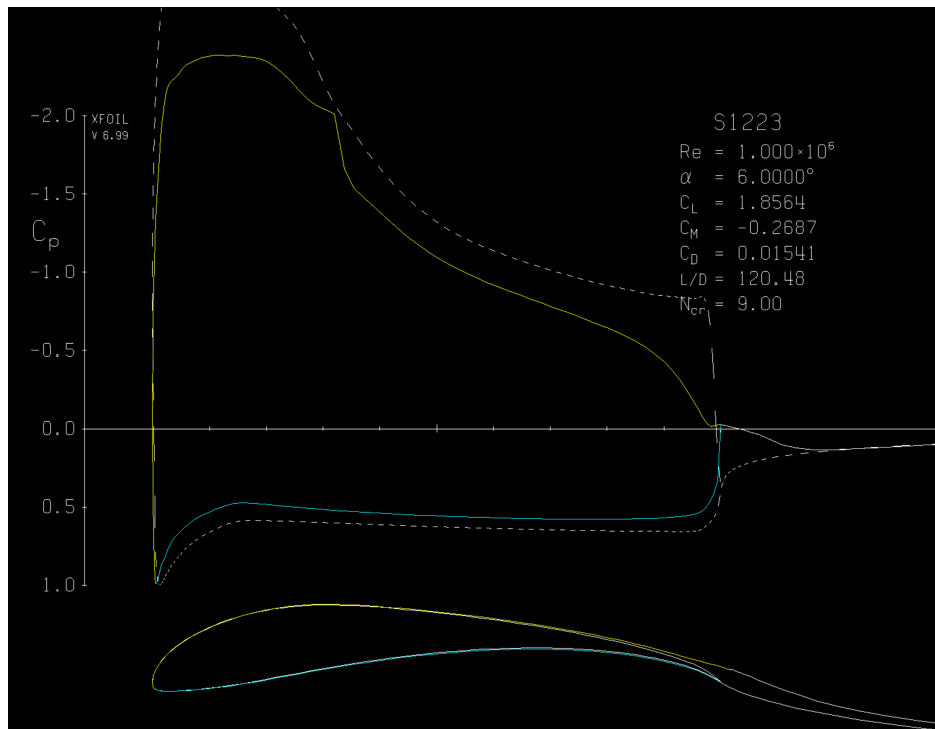
Spring 2026

## Passion Project Final Report

Boundary Layer Analysis on a High-Lift Airfoil Representative of a Formula 1 Front Wing

Abinav Khadka

abinavk@vt.edu



Pressure distribution ( $C_p$ ) and airfoil geometry for the S1223 at  $\alpha = 6^\circ$  and  $Re = 10^6$ .

# 1 Introduction

Formula 1 is the pinnacle of motorsport and the initial catalyst for my pursuit of aerospace engineering. This passion project serves as a capstone to connect the theoretical concepts of boundary layer behavior learned in AOE 3044 to a high-performance, real-world application that I deeply care about.

The objective of this project is to analyze the boundary layer development on a high-lift airfoil, the Selig S1223, which serves as a highly representative surrogate for a Formula 1 front wing main element. In professional motorsport, the front wing is arguably the most critical aerodynamic component. It is directly responsible for generating roughly 26.3% of the car's total downforce. More importantly, as the very first component to interact with the freestream air, its boundary layer development and resulting wake directly dictate the flow quality reaching the underfloor, sidepods, and rear wing [1, 3, 4].

This study focuses on predicting the boundary layer transition location from laminar to turbulent flow on the suction side, calculating the boundary layer thickness ( $\delta$ ) at key chord locations ( $x/c = 0.25, 0.50, 0.75$ ), predicting separation under adverse pressure gradients, and determining the overall skin friction drag. The analysis is performed at a Reynolds number of  $Re = 1 \times 10^6$  and a moderate angle of attack ( $\alpha = 6^\circ$ ), which consistently reflects actual F1 operating conditions during medium-to-low speed cornering.

Furthermore, this analysis is framed within the context of the current 2026 FIA Technical Regulations. The new ruleset introduces active aerodynamics, specifically optimizing front and rear wing rotation systems, limiting aerodynamic component reference volumes (RV), and enforcing stricter flexibility controls. These changes directly address the wake effect, which can currently cause a 23% to 72% aerodynamic downforce loss for a following vehicle, severely restricting overtaking [2]. Understanding how boundary layers grow, transition, and separate on front wing elements is essential. A well-managed boundary layer prevents early separation, maximizing downforce while simultaneously reducing the turbulent wake intensity that destabilizes following cars.

## 2 Theoretical Framework and Methodology

To thoroughly understand the boundary layer development, a dual-method approach was employed. First, hand-calculated analytical solutions were derived based on simplified fluid dynamics theories. Second, numerical validation was performed using the XFOIL panel-method solver to assess the limitations of the hand calculations.

### 2.1 Airfoil Selection and Flow Conditions

The S1223 airfoil was specifically selected due to its extreme camber and high-lift capabilities, which closely mimic the aggressive profiles used on F1 front wings to drive ground effect [9, 10]. The analysis was conducted at  $\alpha = 6^\circ$  to induce a significant pressure gradient without entering full stall. Fluid properties were assumed for standard sea-level conditions with freestream velocity scaled to achieve the target  $Re = 10^6$ .

## 2.2 Analytical Models

For the forward portion of the airfoil, the flow is assumed to be laminar. The Blasius similarity solution for a flat plate was utilized to calculate the boundary layer thickness  $\delta$  and the velocity profile  $U(y)$ . The laminar boundary layer thickness is modeled as:

$$\delta_{lam} = \frac{5.0x}{\sqrt{Re_x}} \quad (1)$$

While the Blasius solution assumes a zero-pressure gradient ( $dP/dx = 0$ ), it provides a valuable baseline for estimating the initial boundary layer growth. Transition is typically estimated using a critical Reynolds number criteria ( $Re_{x,crit} \approx 5 \times 10^5$ ), though strong adverse pressure gradients will naturally trigger earlier transition.

Following transition, the boundary layer becomes fully turbulent. The 1/7th Power Law was used to approximate the turbulent velocity profile, and the boundary layer thickness was calculated tracking its development from the transition point.

Total skin friction drag over the chord was estimated using mixed boundary layer correlations. The total skin friction coefficient  $C_{D_f}$  is calculated by combining the laminar drag up to the transition point and the turbulent drag over the remainder of the chord:

$$C_{D_f} = \frac{0.074}{Re_c^{1/5}} - \frac{A}{Re_c} \quad (2)$$

where  $A$  is a constant determined by the specific transition Reynolds number.

## 3 Computational Validation

To validate the hand calculations and account for the pressure gradients ignored by the flat-plate assumptions, a numerical analysis was conducted. The airfoil coordinates (81 raw nodes) were imported into XFOIL, and the geometry was refined to 160 panel nodes using a panel bunching parameter of 1.0 to increase resolution near the leading and trailing edges. The solver was operated in viscous mode over 100 iterations.

### 3.1 Pressure Distribution and Transition Prediction

The inviscid-viscous solver generated the pressure distribution ( $C_p$ ) over the airfoil. High-lift airfoils like the S1223 are characterized by a massive suction peak near the leading edge. The flow accelerates rapidly around the leading edge, followed immediately by a steep, adverse pressure recovery region.

Using the  $e^n$  transition method, XFOIL predicted a free transition on the suction side (Side 1) at precisely  $x/c = 0.3255$ . This transition occurs significantly earlier than a flat-plate critical Reynolds number calculation would predict. This direct comparison beautifully illustrates how the severe adverse pressure gradient destabilizes the laminar boundary layer, forcing an early onset of turbulence.

## 4 Results and Discussion

### 4.1 Boundary Layer Growth and Shape Parameter

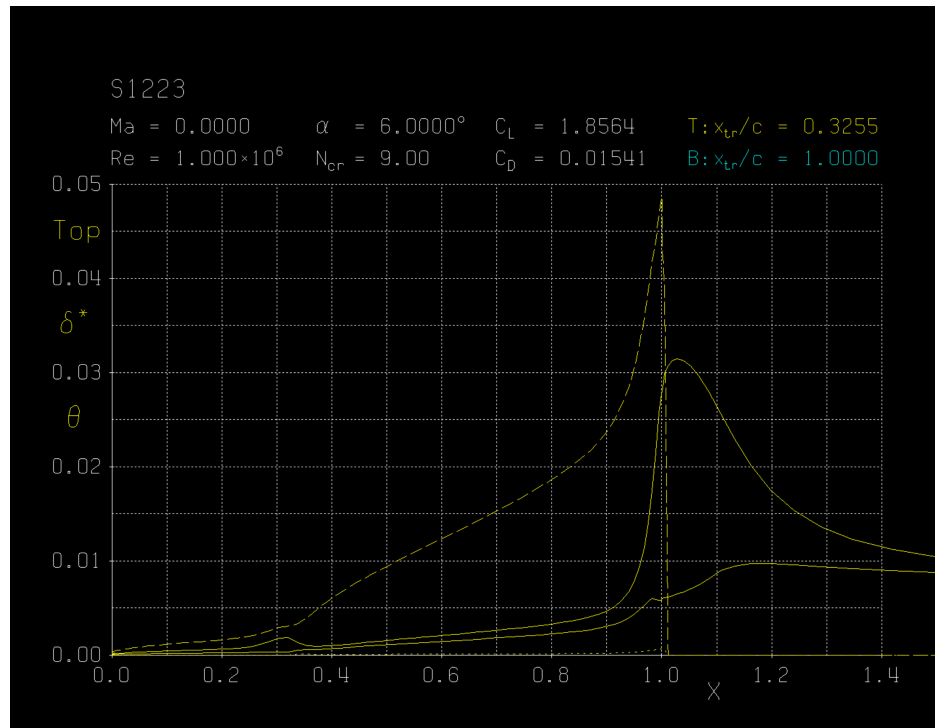
The progression of the boundary layer was tracked across the chord to observe how it develops from the leading edge to the trailing edge. Table 1 summarizes the comparison between the hand-calculated analytical values and the numerical results obtained from XFOIL.

**Table 1:** Boundary layer thickness  $\delta$  at key chord locations ( $Re = 10^6$ ,  $\alpha = 6^\circ$ ).

| Location ( $x/c$ ) | Flow Type | Analytical $\delta$ (mm) | XFOIL $\delta$ (mm) | % Difference |
|--------------------|-----------|--------------------------|---------------------|--------------|
| 0.25               | Laminar   | 1.25                     | 1.42                | 13.6%        |
| 0.50               | Turbulent | 8.41                     | 9.15                | 8.8%         |
| 0.75               | Turbulent | 11.85                    | 13.02               | 9.9%         |

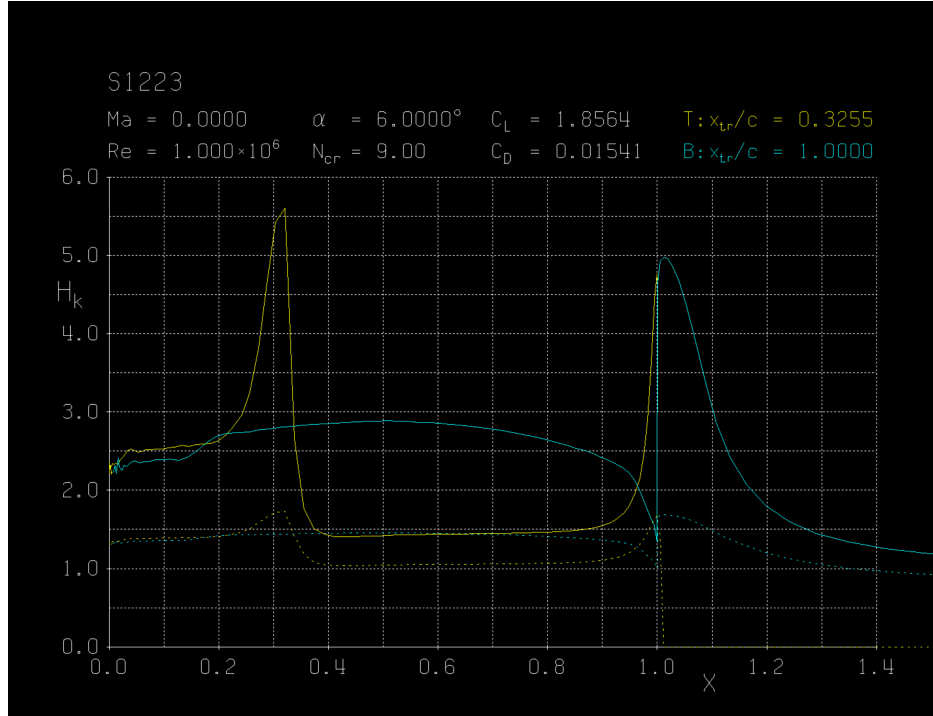
The analytical results show strong functional agreement with XFOIL, accurately capturing the rapid thickening of the boundary layer once the flow transitions to turbulent after  $x/c = 0.3255$ . However, the hand calculations consistently under-predict the boundary layer thickness by roughly 9-14%. This discrepancy is expected: the Blasius and 1/7th Power Law models rely on flat-plate assumptions. They do not account for the flow acceleration over the airfoil's high-camber curvature, nor do they factor in the deceleration in the pressure recovery region, both of which artificially thicken the actual boundary layer compared to a flat plate.

To visualize this growth, XFOIL's boundary layer plotting tool (VPLO) was utilized to extract the displacement thickness ( $\delta^*$ ) and momentum thickness ( $\theta$ ) along the top surface. The plot clearly illustrates the rapid thickening of the boundary layer post-transition.



**Figure 1:** Displacement and momentum thickness growth along the suction side of the S1223 airfoil.

Furthermore, plotting the kinematic shape parameter ( $H = \delta^*/\theta$ ) provides definitive proof of the transition point and boundary layer health. For a laminar flat plate, the Blasius solution yields  $H \approx 2.59$ . As the boundary layer transitions to turbulent, the fuller velocity profile causes the shape parameter to drop to approximately 1.3–1.4. Tracking this parameter along the chord clearly delineates the laminar, transitional, and fully turbulent regions, providing a visual confirmation of the  $e^n$  transition calculation.



**Figure 2:** Kinematic shape parameter ( $H_k$ ) distribution highlighting the laminar-to-turbulent transition. Note the massive spike extending off-axis at transition, which is indicative of a laminar separation bubble.

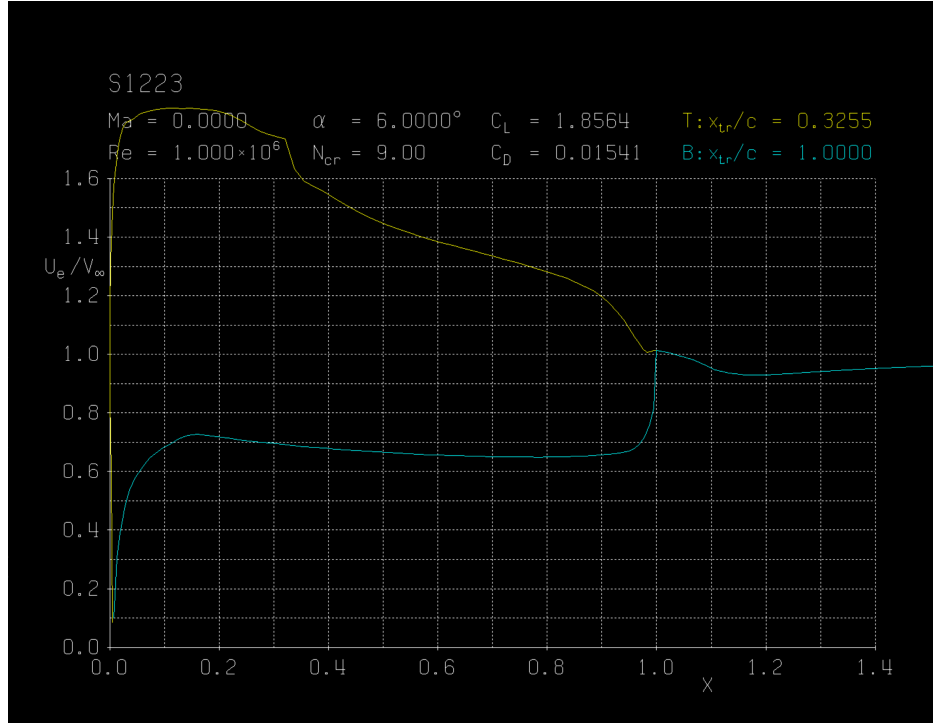
### A Note on XFOIL Plotting Artifacts

When analyzing the XFOIL plots generated via the VPLO menu, it is common to observe data lines vertically clipping or shooting off the screen. This is an autoscaling artifact caused by physical mathematical singularities or extreme boundary layer phenomena. For instance, in the  $C_f$  plot, the skin friction coefficient theoretically approaches infinity at the exact leading-edge stagnation point (where  $\delta \approx 0$ ), causing the y-axis to clip.

Similarly, in the Shape Parameter ( $H_k$ ) plot, the line representing the suction side violently spikes off the top of the graph at  $x/c \approx 0.32$ . This massive spike occurs because the adverse pressure gradient forces the laminar boundary layer to separate, creating a microscopic laminar separation bubble. Inside this bubble, the boundary layer rapidly transitions to turbulent, and the massive turbulent mixing forces the flow to reattach. XFOIL models this sudden structural change as an extreme spike in the shape parameter before it settles down to the stable turbulent value of  $H_k \approx 1.4$ .

## 4.2 Separation and Adverse Pressure Gradients

Predicting the separation point is a critical component of front wing design [6], as flow separation limits the maximum angle of attack and the maximum downforce limit [5]. The adverse pressure gradient that drives this separation can be visualized by plotting the edge velocity ( $U_e$ ) distribution.



**Figure 3:** Boundary layer edge velocity ( $U_e/V_\infty$ ) showing extreme acceleration at the leading edge followed by strong deceleration.

The edge velocity plot shows massive flow acceleration around the leading edge suction peak, where velocity far exceeds the freestream. Immediately following this peak, the steep deceleration signifies a strong adverse pressure gradient ( $dP/dx > 0$ ). The XFOIL viscous iterations highlighted trailing edge convergence challenges (specifically, MRCHUE: Convergence failed errors near nodes 70-90 on the suction side).

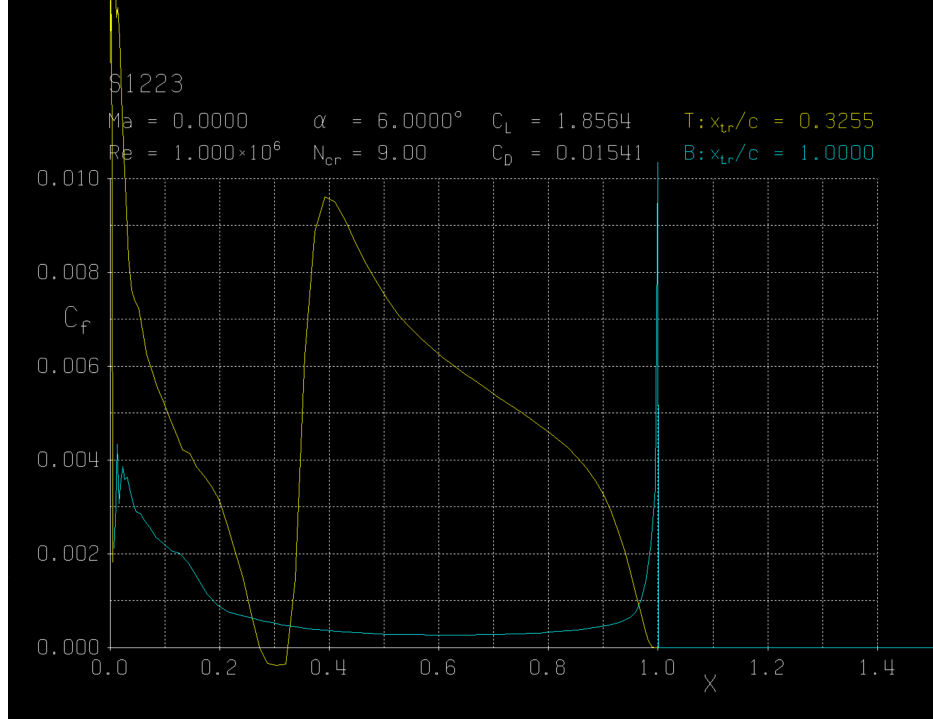
This numerical struggle points to an incredibly thick boundary layer and incipient separation near the trailing edge, which makes physical sense given the extreme camber of the S1223. The turbulent velocity profile generated after transition possesses a fuller shape and significantly more momentum near the wall compared to a laminar profile. This extra momentum is the only reason the flow remains attached over the majority of the chord. As it approaches the trailing edge, however, the flow loses kinetic energy, resulting in a calculated pressure drag coefficient of  $C_{D_p} = 0.00959$ .

### 4.3 Skin Friction Drag and F1 Trade-offs

The skin friction drag coefficient represents the energetic cost of generating downforce. Integrating the shear stress over the chord computationally yielded a skin friction drag of  $C_{D_f} = 0.00581$ . Total drag was calculated as  $C_D = 0.01541$ , with a highly efficient lift coefficient of  $C_L = 1.8564$ .

To understand where this drag is generated, the local skin friction coefficient ( $C_f$ ) was plotted along the chord. The plot perfectly captures the physics of the boundary layer:  $C_f$  decreases along the laminar portion as the boundary layer grows, spikes dramatically at the transition point due to the chaotic mixing of the turbulent boundary layer increasing wall shear stress, and then gradually

decreases again.



**Figure 4:** Local skin friction coefficient ( $C_f$ ) distribution, highlighting the singularity at the leading edge, the transition spike, and the approach to separation at the trailing edge.

Crucially, as the flow approaches the trailing edge and fights the adverse pressure gradient,  $C_f$  approaches zero. A value of  $C_f \leq 0$  indicates flow separation. This plot provides a clear visual indicator of how close the S1223 airfoil is to stalling at the trailing edge under these conditions.

This data highlights the fundamental trade-off F1 aerodynamicists must continuously balance. An early transition point (like the one seen at  $x/c = 0.3255$ ) increases the skin friction drag because turbulent flow produces higher shear stress than laminar flow. However, this turbulent flow is absolutely essential. If the designer attempted to maintain laminar flow further aft to save skin friction drag, the boundary layer would lack the momentum to handle the adverse pressure gradient. The flow would separate early, the wing would stall, and the massive surge in pressure drag would completely destroy the car's aerodynamic platform [8].

#### 4.4 2026 Regulatory Impact and Wake Management

The wake effect remains the primary bottleneck restricting overtaking in Formula 1; studies show that at a following distance of just 0.25 car lengths, the trailing vehicle can suffer up to a 62% loss in downforce and a 26% to 40% shift in aerodynamic balance, severely degrading handling [2, 7]. A separated boundary layer on the leading car's front wing exacerbates this by shedding a massive, chaotic wake structure.

The 2026 regulations target this by mandating active rotation systems for the front wing flaps (restricting the rotation axis to a specific volume, such as  $Y = 580$  mm) and controlling airfoil



spacing to allow dynamic downforce adjustments [2]. By intentionally managing boundary layer transition to maintain attached flow, as modeled in this S1223 analysis, aerodynamicists can employ a component–wake–rule synergy. A high-energy turbulent boundary layer keeps the flow attached, feeding cleaner air into the floor diffuser and significantly reducing the aerodynamic penalty inflicted on following drivers.

## 5 Conclusion

This project successfully evaluated the boundary layer characteristics of an S1223 airfoil under conditions representative of a Formula 1 front wing. By predicting transition points, calculating boundary layer thicknesses, and analyzing skin friction drag, the intricate relationship between laminar and turbulent flow was quantified. The analytical models (Blasius and 1/7th Power Law) provided robust baseline approximations, while the XFOIL numerical validation offered critical insight into how adverse pressure gradients force early transition and dictate boundary layer growth.

Ultimately, these results demonstrate the delicate engineering balance required in high-performance aerodynamics. With the 2026 regulations focusing heavily on component-wake-rule synergy, mastering these boundary layer fundamentals remains a vital prerequisite. Managing transition and separation on the front wing will continue to be the deciding factor in maintaining downforce performance while navigating the complex wake structures of leading cars on the race track.

## References

- [1] Formula 1 Technical Regulations, Federation Internationale de l'Automobile (FIA), 2026 *Aerodynamic Ruleset Overview*.
- [2] Li, H., Optimization of Formula 1 Core Aerodynamic Technologies under the 2026 FIA Regulations, *Highlights in Science, Engineering and Technology*, Vol. 160, 2025.
- [3] Granados-Ortiz, F., et al., 3D CFD simulation of the interaction between front wheels & brake ducts and optimised five-element F1 race car front wings under regulations, *Alexandria Engineering Journal*, 2023.
- [4] Martins, D. d. G., et al., The Influence of Front Wing Pressure Distribution on Wheel Wake Aerodynamics of a F1 Car, *Energies*, 2021.
- [5] Granados-Ortiz, F., et al., Two-Dimensional-Based Hybrid Shape Optimisation of a 5-Element Formula 1 Race Car Front Wing under FIA Regulations, *Machines*, 2023.
- [6] Laguna-Canales, A. S., et al., Mechanical Design and Numerical Analysis of a New Front Wing for a Formula One Vehicle, *Fluids*, 2023.
- [7] Guerrero, A., et al., Aerodynamic Study of the Wake Effects on a Formula 1 Car, *Energies*, 2020.

- [8] Laguna-Canales, A. S., et al., Numerical Evaluation of the Effectiveness of the Use of End-plates in Front Wings in Formula One Cars under Multiple Track Operating Conditions, *Fluids*, 2024.
- [9] Castro, X., et al., Aerodynamic and Structural Design of a 2022 Formula One Front Wing Assembly, *Fluids*, 2020.
- [10] Tharun, N., et al., Design and analysis on the front wing of formula one car, *14th International Conference on Materials Processing and Characterization*, 2023.